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SUMMARY

MS Computer Science student at Arizona State University with practical experience in building full-stack systems from backend APIs and distributed databases to blockchain applications. Built an end-to-end blockchain fraud detection system (Solidity smart contracts, Ethereum, ML anomaly detection) and a production-ready REST API on a geo-distributed database across multiple AWS regions. Strong knowledge of software engineering principles, designing systems with security considerations and delivering working code autonomously.

EDUCATION

Arizona State University, USA *Aug 2025 - Present*
Master, Computer Science — Tempe, AZ | GPA: 3.44

Velammal Institute of Technology, India *Jun 2019 - Apr 2023*
Bachelors, CSE — Chennai, India

SKILLS

- **Languages & Backend**: Python, JavaScript, SQL, FastAPI, Flask, Node.js, REST API design
- **Blockchain & Web3**: Solidity, Hardhat/Ethereum, smart contract development, NFT minting workflows
- **Databases & Systems:** CockroachDB , MongoDB, SQLite, PostgreSQL
- **Machine Learning**: Isolation Forest anomaly detection, scikit-learn, model integration into production APIs
- **Frontend & Visualization**: JavaScript, Next.js, D3.js, HTML/CSS
- **Tools**: Git, GitHub, VS Code, AWS

PROJECTS

VeridiumMesh — Blockchain-Based Carbon Credit Fraud Detection System | *Python, FastAPI, Solidity, Hardhat/Ethereum, Next.js, Isolation Forest Github*

- Built an end-to-end ML + blockchain system combining an Isolation Forest anomaly-detection model, a FastAPI backend, Solidity smart contracts deployed to a local Ethereum network, and a Next.js frontend to screen carbon credits before minting them as NFTs.
- Developed backend audit and history-tracking functions and configured endorsement policies governing the credit lifecycle (submit, score, approve, mint, retire)

FreshCart — Geo-Distributed Real-Time Inventory Management System — *CockroachDB Cloud, Node.js, Express, JavaScript, REST API, AWS* | GitHub

- Built a real-time dashboard connected to a geo-distributed CockroachDB Cloud cluster across 4 AWS regions, visualizing live cluster health, regional analytics, and product search/filtering for 5,045+ products
- Designed database schema with strategic composite indexes achieving 65-75% query latency reduction vs. full table scans
- Deployed a production-grade RESTful API (7 endpoints, full CRUD) with connection pooling, pagination, filtering, and SSL-secured connectivity on AWS us-east-1
- Validated system performance through custom test suites — 49ms average read latency, 14.07 QPS throughput, and 100% availability across 20 fault-tolerance queries with zero data loss

Anti-Surveillance Data Visualization — Parallel Coordinates Plot — *D3.js, Figma, GitHub* | GitHub

- Designed and implemented an interactive parallel coordinates visualization exploring the anti-surveillance perspective of a CCTV & crime dataset, encoding each city as a polyline across CCTV per capita, crime index, population, and total camera count
- Built hover and filtering interactions to let users compare cities and surface tradeoffs between surveillance density and crime outcomes

CERTIFICATIONS

- Google Data Analytics Professional Certificate: Coursera
- Python for Data Science: IBM
- SQL for Data Analysis: Udemy